

Integration of a Shape Memory Alloy with a Soft Pneumatic Actuator to Improve the Haptic Interaction Performance of a Soft Social Robot

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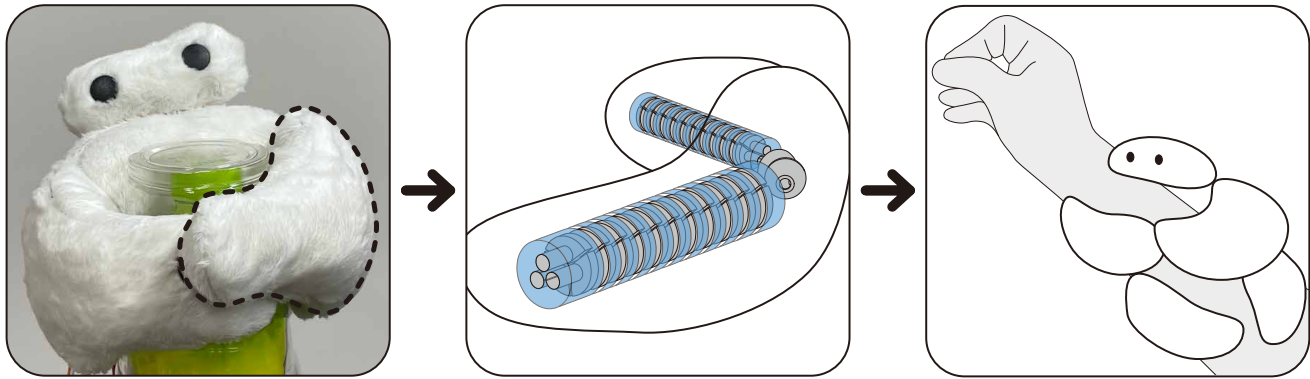


Figure 1: Integration of hybrid actuators into Reliebo [98, 99], a soft robot companion. (Left) The external appearance of Reliebo emphasizing its comfortable and friendly design. (Middle) Conceptual illustration of the arm segment of Reliebo equipped with the proposed hybrid actuator. (Right) A future version of Reliebo with the hybrid actuator, demonstrating the potential for enhanced engagement and interactivity with users through direct physical contact.

ABSTRACT

This paper discusses the integration of a soft pneumatic actuator (SPA) with a shape memory alloy (SMA) to improve haptic technologies in social robotics. The use of traditional rigid actuators in robots can lead to unnatural interactions, while SPAs offer flexibility but may lack precision. SMAs are known for their shape memory effect and precision, but they also have limitations, such as time-consuming heating-cooling cycles. Our research proposes a hybrid actuator that combines the adaptability of SPAs with the precision of SMAs to enhance haptic interaction with social robots. The hybrid actuator is designed to overcome the individual limitations of SPAs and SMAs and achieve a synergistic effect. As a first attempt, the hybrid actuator is intended for use in Reliebo, an inflatable social robot, to provide more natural and responsive interactions. This approach has the potential to improve the emotional and haptic aspects of human-computer interaction and social robotics.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; *Interaction design process and methods*; • **Hardware** → **Sensors and actuators**.

KEYWORDS

Soft Pneumatic Actuator, haptic device, soft robotics, social touch

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1 INTRODUCTION

Social robots play an increasingly important role in modern society [39, 55, 67]. They are utilized in various fields, including education [21, 25, 81–84], healthcare [5, 15, 65, 66, 71, 88, 89, 98], entertainment [18, 29, 34], and daily life [28, 53, 68], to interact with humans through verbal communication [4, 52, 72], visual motion [16, 40, 46], and direct touch [15, 87, 90, 98, 100]. Haptic interaction is particularly important [74, 80, 91, 100] for facilitating both one-way [14, 23, 90] and two-way [74, 75] communication. Traditionally, rigid actuators such as motors and linear actuators have been used for this purpose [19]. However, these conventional

actuators often lead to unnatural interactions and are restricted to predetermined touches [101, 103].

There is an increasing interest in soft actuators composed of flexible or elastic materials. These materials offer a significant contrast in motion flexibility compared to their rigid counterparts [70, 85, 92]. Soft actuators' elastic nature enables them to adapt well to external environments, making them increasingly popular in soft robotics and human-computer interaction due to their safety and versatility [2, 7, 24, 37, 38, 58, 70, 78]. Among the various technologies studied for actuating elastic structures, Soft Pneumatic Actuators (SPAs) stand out due to their lightweight design and simple implementation [9, 12, 17, 30, 36, 42, 44, 48, 49, 57]. SPAs contain enclosed chambers and use controlled positive or negative fluid pressure for actuation [30, 48, 92]. A flexible and smooth SPA, capable of generating forces comparable to rigid actuators, is expected to maximise the effectiveness of social touch and enhance emotional interactions.

However, the SPA has certain limitations, such as slow response times and a lack of precision and repeatability. To mitigate these issues, a Shape Memory Alloy (SMA) could be proposed. SMAs are also widely used in the fields of soft robotics [8, 37, 41, 76] and Human-Computer Interaction (HCI) [43, 56, 77, 96], offering high energy density and precise control. Despite its advantages, the heating and cooling cycles required for actuation are time-consuming. Additionally, the actuator is limited to predefined shape changes, which poses challenges in generating diverse motions [22, 33, 43, 45, 51, 62, 63]. To address these issues, we have developed a fused hybrid actuator that aims to compensate for the shortcomings of each technology and achieve synergistic effects in their operation.

The aim of this research is to improve the structural stability of existing SPA systems, and enhance emotional and affective touch interactions in Reliebo [98, 99], a hand-held wearable inflatable robot (Fig. 1), by developing a hybrid actuator that integrates SPA with SMA. The potential of this hybrid actuator extends to a wide range of inflatable devices [1, 26, 32, 54, 69], making it crucial not only for Reliebo but also for other applications. For instance, our suggested hybrid actuator can reproduce of emotional and affective touches, such as stroking an arm or providing a firm grip. This can be achieved by utilizing the touch provided by conventional single SPA or single SMA configurations. The hybrid actuator offers a wider range of intricate movements with a gentle and natural touch, representing a significant improvement in haptic interactions with social robots and beyond. This is particularly relevant in applications that use inflatable structures.

2 RELATED WORK

2.1 Existing Research on SPA

SPA has gained significant attention in the fields of soft robotics and human-computer interaction due to its ability to provide flexible movements and adapt elastically to the external environment, in contrast to traditional rigid materials. Although there are soft actuators that do not rely on pneumatics [7, 24, 37, 58], the integration of pneumatics offers distinct advantages such as lighter weight and simpler implementation. Therefore, they are currently a subject of active research in the field of soft actuators [85, 93].

Common materials used in SPAs, similar to other soft actuators, are primarily elastic materials. These flexible materials form structures that enclose sealed chambers within [9, 12, 17, 30, 47]. Actuation in an SPA is controlled by the application of positive or negative pressure to fluids (either gases or liquids) within these chambers [30]. The response of the actuator to the applied fluid pressure is determined by different patterns of the chambers, resulting in different movements such as bending, stretching, or twisting. The design of these chambers, including their shape and the materials used, can significantly affect the force exerted and the speed at which the actuator operates [47, 78, 92].

Traditional manufacturing methods for SPA typically involve hand molding soft silicone rubbers such as Ecoflex or DragonSkin into 3D printed molds [27, 47, 65, 78]. Silicone rubber-based SPA offers a high degree of deformation, making it ideal for direct human interaction. The manufacturing process allows for the integration of sensors or other actuators, enabling the creation of complex actuator systems [48, 93, 102]. However, this manual approach often results in reduced accuracy, and the need for high-resolution 3D printers to produce the molds presents its own set of challenges [92]. To overcome these limitations, research has explored the production of soft actuators using 3D printing techniques [78, 95].

A notable structural archetype in SPA development is the bellows structure [9, 10, 12, 30, 86]. Actuators using this structure have been developed primarily for applications such as soft grippers [27, 85, 97] or as assistive devices for bending actions in hands or arms [12, 57, 73]. In addition, SPA has been used to create user-interactive interfaces, as exemplified by developments such as MorphIO [48]. These interfaces demonstrate the versatile application potential of SPA in interactive technology.

2.2 Existing Research on the Integrated SMA and Soft Materials and Their Use in Haptic Presentation

SMAs are unique materials that remember their original shape and return to it when heated after being deformed [3]. This property, known as shape memory effect, is due to the ability of materials to undergo phase transitions at certain temperatures. The most common type of SMA, Nitinol, a nickel-titanium alloy, is best known for its high energy density and precise control [61]. SMAs have been widely explored for their potential in various applications, including haptic and medical devices, due to their ability to provide significant force and precise motion in compact shapes [13, 43, 64, 76, 79, 94].

The integration of SMA with soft materials has opened new possibilities for creating actuators that combine the high energy density and precision of SMA with the flexibility and safety of soft materials [31, 35, 50, 76]. Recent research has focused on embedding SMA wires or springs in soft, elastomeric matrices. This hybrid approach enables the creation of actuators that can perform complex, biomimetic motions, ideal for applications that require gentle yet precise interactions. For example, SMA-infused soft actuators have been used to mimic the motion of natural organisms, offering new solutions in soft robotics for tasks that require delicate handling or adaptability to changing environments.

SMAs are increasingly used in haptic feedback and tactile display technologies [6, 43, 56, 77]. Their ability to rapidly change shape and

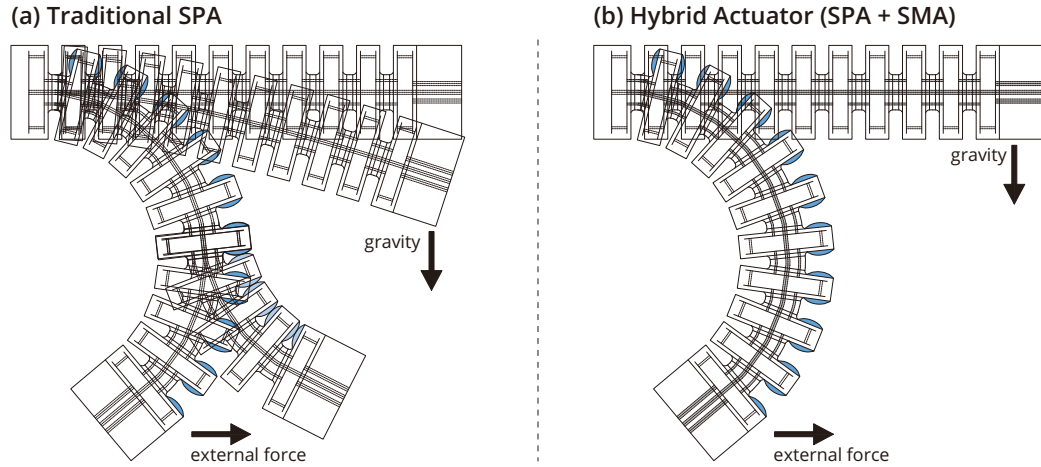


Figure 2: Comparative illustrations of actuator response to gravity and external forces. (a) Traditional SPA: Under the influence of gravity, the inherent weight of the elastic material causes the SPA to naturally bend downward. When pressurized, the SPA bends in the opposite direction of the expanded chamber (shown with the chamber colored in blue), a reaction known as bending actuation. However, if an external force is applied in the opposite direction of this bending actuation, the structure may collapse, causing in the actuator to bend in the direction of the applied force. **(b) Hybrid actuator (SPA + SMA):** In its neutral state, the SMA remains unheated within the three chambers, maintaining martensitic form and allowing the actuator to maintain an upright position against gravity. When heat is applied via electrical current, the SMA transitions to its austenitic phase, ensuring that the actuator remains upright and resists bending due to gravity. During bending actuation caused by chamber expansion, only the SMA on the inside of the bend transitions to austenite. This selective phase transformation ensures that the bending structure of the actuator remains intact even when external forces are applied in the opposite direction of bending. In the figure, the actuator shows the expansion of a single chamber (highlighted in blue), with the integrated SMA within the hybrid actuator shown in gray.

stiffness makes them suitable for simulating different textures and touch sensations. Research in this area has led to the development of wearable haptic interfaces and tactile displays that use SMA actuators to generate dynamic, responsive touch sensations. These devices can simulate different surface textures or provide directional cues to enhance the user experience in virtual reality or provide critical feedback in remote or minimally invasive surgery. The combination of SMA's precise control and the nuanced feedback it can provide has made it a promising tool for advancing haptic technology in various fields [20, 33].

3 OUR APPROACH

3.1 Integration of SPA and SMA in Actuator Design

The theoretical integration of SPA with SMA is expected to leverage the unique strengths of each while addressing their individual limitations. Although this concept has not been extensively researched, it has the potential to synergistically enhance the capabilities of both technologies.

SPA's inherent advantage in facilitating flexible motion and adaptability to external stimuli has catalyzed its prominence in soft robotics and human-computer interaction. However, a notable limitation of SPA is its susceptibility to compromise in neutral states in the absence of pressure, which limits its range of applications.

Conversely, SMA is characterized by its intrinsic ability to memorize and return to its preformed shape upon thermal activation. This property, when combined with SPA, enhances the structural integrity of the actuator in its neutral state, compensating for the intrinsic shortcoming of SPA.

In addition, the integration of SMA within the SPA framework promises improved operational control in a variety of environments. While the functionality of SMA is typically limited to predefined trajectories, the fusion with the adaptive nature of SPA allows for more refined and reactive haptic engagements. For example, in tasks such as encircling a limb, the adaptability of SPA enhances the precision of SMA, resulting in actuators capable of providing consistent haptic feedback under varying external conditions (Fig. 2).

3.2 SPA Model Design and Fabrication

Several configurations are possible when designing SPA from elastic materials. For integration with SMA, this study adopts an omnidirectional bellows structure with three degrees of freedom (DoFs). In previous research, bellows-structured SPAs have been mainly categorized into single and parallel structures [30, 48, 92]. The hybrid actuator proposed in this research is adaptable to both configurations. However, with future integration into social robots in mind, we have chosen a single structure design. This design is characterized by internal chambers divided into three distinct sections. Building on previous work that used finite element modeling (FEM)

to validate the performance of 3D printed SPA, our research references these models for design inspiration [92]. Instead of relying solely on pneumatic control, the goal is to develop a hybrid actuator that incorporates SMA. Therefore, we have slightly modified the internal structure to accommodate SMA annealed into a spring shape within the pneumatic chambers. In addition, the sealed structure of the conventional SPA posed a challenge in anchoring the SMA at both ends. To overcome this, we designed additional end caps to ensure secure integration of both actuators (Fig. 3).

For the construction of the SPA, we used a Form 3+ stereolithography (SLA) printer (Formlabs, Somerville, USA) with Elastic 50A as the material of choice. This material, similar to that used in previous studies, is known for its exceptional flexibility and has the lowest Shore hardness of comparable filaments. The total printing time, including support filaments, was approximately 19 hours. This time allowed four SPAs to be printed simultaneously for testing purposes. After printing, the SPAs underwent a cleaning process that included immersion in isopropyl alcohol for 20 minutes, followed by a curing phase at 60 degrees Celsius for the same duration. We used Form Wash for the cleaning phase and Form Cure for the curing process, both products from the same manufacturer (Formlabs, Somerville, USA). Notably, residual uncured resin remained in the internal chambers of the SPA. To address this, we implemented a methodical process of introducing and then removing cleaning alcohol into these chambers to ensure thorough removal of any remaining uncured material.

3.3 SMA Annealing for Shape Memory Retention

The SMAs used in the production of the hybrid actuator are made of NiTi wire with a diameter of 0.4 mm, and the transformation temperature after heat treatment is 70 degrees Celsius. In the annealing process, the SMA is fixed into a thin spring shape that fits within the diameter of the chamber of the SPA, and then subjected to heat treatment in a mini electric furnace, mini-BS1 (Nitto Science), at 400 degrees Celsius for 30 minutes. During the integration process, the SMA must be securely fixed in a spring mold to fit into the internal chamber of the SPA. The design of the SPA allows for a cylindrical space with a diameter that can accommodate a spring structure up to 2.462mm (Fig. 3). If the force exerted by the SMA is found to be insufficient after initial testing, the diameter of the space designated for the SMA is gradually increased to accommodate larger springs, thereby increasing the force output of the actuator through subsequent test iterations. After annealing, the ends of the SMA are connected to wires to allow for deformation by heating induced by the application of voltage.

4 PROPOSED TESTING METHODOLOGY AND EVALUATION CRITERIA

4.1 Description of the Testing Setup

Figure 4 shows the setup for testing the hybrid actuator that integrates both SPA and SMA. The actuator is controlled by an Arduino Mega 2560 microcontroller. The pneumatic pressure applied to the actuator is generated by an air compressor (SR-102, Fujiwara Industrial, Japan), which can reach up to 0.8 MPa. However, such high

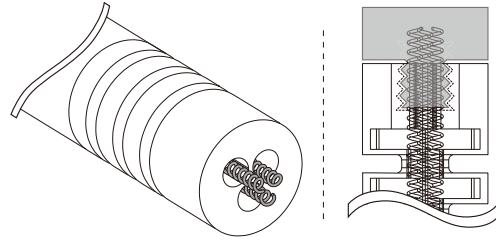


Figure 3: Illustration of SMA integration into an SPA to form a hybrid actuator. The left segment of the figure shows the three chambers of the SPA, each containing a spring-loaded SMA (shown in gray) that is designed to fit within the confines of the SPA structure. The right segment shows a redesigned lid for the traditional sealed SPA (shown in gray), which facilitates the encapsulation of the SPA after SMA installation.

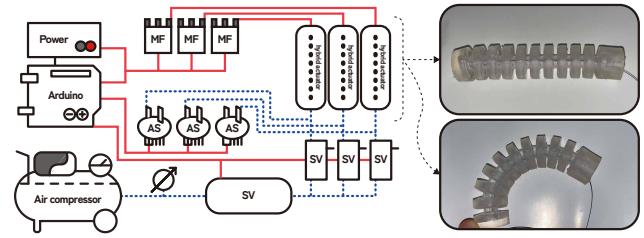


Figure 4: Experimental setup to evaluate the performance of a hybrid actuator. The red solid lines represent electrical wiring and the blue dashed lines represent silicone tubing carrying compressed air. The hybrid actuator was designed as an integrated structure, but each chamber is shown separately in this figure for clarity. The black dashed lines within the chambers represent SMAs. The right side of the figure shows an SMA inserted into the actual fabricated SPA (before annealing). The elastic material SPA will reproduce the bending behavior shown in the lower left when air is injected (manually bent in the photo).

pressure is not required for actuator operation, so a flow regulator is used to adjust both flow and pressure, followed by a solenoid valve (labeled SV in Figure 4) to control the ON/OFF output of the air pressure. Since the actuator contains three chambers, individual solenoid valves with ports 1 and 2 and 2 and 3 are connected to the inlet of each chamber for air injection. In addition, T-fittings are provided to allow the attachment of air pressure sensors (labeled AS in Figure 4) to monitor the air pressure within the chambers. To activate the SMA integrated in the hybrid actuator (heating by electric current), a MOSFET circuit (labeled MF in Figure 4) is connected to the SMA, which controls the ON/OFF of the voltage application and is then controlled by the Arduino Mega. The required voltage and current levels are initially estimated based on previous research [43], but they vary depending on the diameter and length of the SMA used. Therefore, the actuator will be tested with increasing voltage levels to determine the most efficient and safe heating times to operating temperature.

4.2 Proposed Testing Procedures

The hybrid actuator will be tested under the following conditions within the established test environment.

Tests for Support Strength in Neutral State.

- (1) Verification of the ability of the actuator to maintain a stationary position in its neutral state at 30 degree intervals from a vertical orientation to parallel to the ground.
- (2) Evaluate the ability of the actuator to adjust in the direction of yaw (relative to the ground, regardless of the tilt of the actuator) from the point of ground attachment.

Tests for Grip Strength in Bending State.

- (1) Evaluation of grip strength when holding bars of various diameters (pressure sensors are attached to the bars, not the actuator, to measure the force applied).
- (2) Measure resistance to external forces when the actuator is oriented vertically and parallel to the ground.

5 DISCUSSION AND FUTURE WORK

We are exploring a new approach by integrating SPA with SMA to create a hybrid model. This approach is expected to improve actuator technology by combining the adaptability and flexibility of SPA with the precision and energy efficiency of SMA. This combination is expected to expand the potential applications of actuators, especially in the fields of social robotics and human-computer interaction.

Based on an omnidirectional bellows structure SPA with three internal chambers, we used a 3D printed SPA made of elastic filaments. An SMA annealed into a spring shape was integrated into the internal chambers of the SPA to form a robust hybrid actuator. The integration is expected to significantly improve the structural integrity of the actuator in its neutral state, potentially addressing a key limitation of conventional SPAs. Such an improvement is expected to be critical to ensuring consistent performance under varying environmental conditions in future tests. The hybrid actuator's dynamic range of motion, made possible by complementing the pre-programmed motions of the SMA with the flexibility of the SPA, enables customized applications in various fields, including personalized assistive devices.

Although our proposed hybrid actuator is promising, it has certain limitations. One of the limitations is the issue of power consumption within the hybrid actuator. Although the SMA installed in the actuator can exert significant force at low power, making it highly efficient in terms of power-to-force ratio, the power efficiency for generating the necessary air pressure in the SPA remains a challenge. To achieve sufficient air pressure, we initially plan to use air compressors and solenoid valves under experimental conditions. However, for future development of actuators suitable for wearable devices, it is imperative to optimize the pneumatic system. For example, instead of using air compressors to provide air pressure, one could consider using gas cylinders filled with compressed air [11], or replacing the solenoid valves used to seal the internal air of the hybrid actuator with valves that do not require electricity [59, 60], or alternatively, controlling the opening and closing of the air supply and exhaust ports using SMAs. Traditionally, the SMA is

embedded in silicone molds during the molding process. In our approach, the SMA is inserted into the cavities of the 3D printed SPA, which ensures consistent placement, but faces challenges due to the irregular fabrication and annealing process of the SMA. Future work will include custom SMA springs for quantitative evaluation of the actuators. In addition, while our current focus is on large-scale designs, future efforts will focus on miniaturizing the actuators for haptic interactions in social robots. Careful consideration will be given to ensuring that the force applied during interactions is ethically appropriate. Incorporating the hybrid actuator into social robots involves generating motion by both pneumatic and electrical means. We plan to introduce an improved "squeeze" haptic by adding an external airbag for a more intuitive haptic representation. These airbags could also serve as pressure sensors after actuator actuation. We also plan to experimentally test grip strength in different shapes and sizes to ensure the suitability of the actuators for different body positions and interaction scenarios. As we are currently in the design phase, the expected effects remain theoretical. Therefore, it is essential to conduct user studies to evaluate the tactile feedback proposed by the hybrid actuators.

6 CONCLUSION

Our hybrid actuator opens avenues for more natural and responsive interactions in social robots, potentially transforming their perception and increasing their adoption in fields such as healthcare and education. This research, while still in its early stages, shows promise for advancing actuator technology, particularly in enhancing the emotional and haptic dimensions of human-computer interaction. However, it must be acknowledged that the current iteration of our hybrid actuator still falls short in terms of overall efficiency when compared to traditional rigid and wire-pulled actuators. Its reliance on air compressors or pumps for pneumatic operation is a limitation, especially from a wearable technology perspective. We believe there is potential for future improvements. By improving the materials used in the SPA to increase inflation efficiency and optimizing the minimum power requirements to operate the SMA, we could eventually achieve multiple activations with minimal power and compressed air supplies. In addition, as conventional rigid actuators continue to advance in miniaturization and weight reduction, the relative advantage of the SPA's light weight may diminish. However, by using pneumatic actuators fabricated with elastic TPU films instead of the elastomeric materials used in this study, we could construct actuation systems using only the weight of the TPU film, pneumatic connector tubing, and lightweight SMA. This approach could potentially balance the trade-offs between efficiency and flexibility.

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